

Beyond Expected Goals: A Probabilistic Framework for Shot Occurrences in Soccer

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What Does It Mean to “Deserve” a Win?

- **Post-match analysis:** We often hear “this team didn’t deserve that result.” But what does this really mean?
- **Intuition:** Deserving refers to the difference in *goal-scoring danger* created or conceded by the two teams.
- **Mainstream approach:** Compare each team’s expected goals (xG)!
- **Our question:** Does xG really capture “deserving”?

Example 1: Comparing Chances

- **Match:** Manchester City vs. Real Madrid (Feb 19, 2025)
- **Key question:** Which chance is more likely to produce a goal?

Example 2: Rebounded Shots

- **Match:** Orlando City vs. Philadelphia Union (Feb 22, 2025)
- **Key question:** How does xG measure the value of this sequence?

Example 2: Rebounded Shots

Time	Player	Shot Outcome (xG)
78:01	Brekalo	Shot Blocked (0.05)
78:02	Muriel	Shot Post (0.52)
78:04	Pasalic	Shot Post (0.68)
78:05	Pasalic	Shot Goal (0.38)

Table: Four shots in one possession totaling 1.63 xG, but only one goal is possible.

Example 3: Mbappé's Brilliance

- **Match:** Manchester City vs. Real Madrid (Feb 19, 2025)
- **Key questions:** When is xG credited? What does it value?

What Are Expected Goals (xG)?

- **xG:** Probability that an *attempted shot* is scored, based on context
 - Distance, angle, shot type, pressure, player positions, etc.
 - Often fitted using ML models (e.g., XGBoost) on historical shot data
- **Purposes:**
 - Shot quality estimation
 - Team performance summaries
 - Finishing or goalkeeping “skill” via residuals (xGOE)

The Problem: xG Answers the Wrong Question

- As constructed, xG models are trained only on *observed* shots.
- So xG answers the following question:

$$\mathbb{P}(\text{goal} \mid \text{shot})$$

- But as we saw, many dangerous moments are *also* about:

$$\mathbb{P}(\text{shot})$$

- What are the consequences of ignoring this?
 - No-shot danger gets 0 credit (Example 1)
 - Rebound sequences inflate summed xG (Example 2)
 - The skill of shot *creation* is under-credited (Example 3)

Modeling the *True* Goal-Scoring Process

- Scoring can be thought of as a two-step process:
 - ➊ **Will a shot happen?** (creating a chance)
 - ➋ **If it happens, how likely is it to be a goal?** (shot quality)
- **We should model both.**
 - xS : the probability a shot occurs in the next second
 - xG : the probability of a goal given a shot occurs
- Putting them together produces **$xG+$** : the probability of scoring in the next second.

Introducing Our New Metric: xG+

- At each time point t , define:

$$\begin{aligned} \text{xG+}_t &= \mathbb{P}_t(\text{goal scored}) \\ &= \mathbb{P}_t(\text{goal} \mid \text{shot}) \cdot \mathbb{P}_t(\text{shot}) \\ &= \text{xG}_t \cdot \text{xS}_t \end{aligned}$$

- Possession-level aggregation (over n time points):**

$$\text{xG+}_{\text{poss}} = 1 - \prod_{t=1}^n (1 - \text{xG+}_t)$$

- What this gives us: the probability of scoring (at most once) in the possession.
- Respects the fundamental limit of at most one goal per possession.

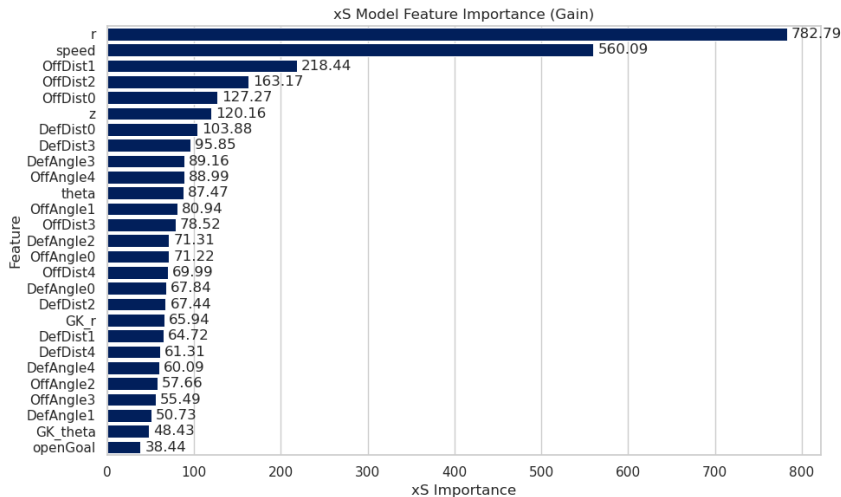
- **Source:** Gradient Sports tracking and event data (EPL 2022–2025)
- **Resolution:** 30 frames per second
- **Inputs:**
 - Player positions (x,y), ball position (x,y,z)
 - Possession indicators, shot events and outcomes
 - Player/team IDs

- Keep frames where the ball is in play and a team has *clear possession* in the attacking third
- Define attacking sequences:
 - Start: possession gained in attacking third
 - End: the defense regains possession or the ball exits the attacking third
- Interpolate ball positions to fill in missing frames

- **Ball details:** distance to goal, angle, height, speed
- **Local pressure:** nearest attackers and defenders relative to ball
- **GK context:** keeper location relative to goal
- **Goal openness (openGoal):**
 - Model defenders as circles (diameter 0.75 m)
 - Compute fraction of goalmouth left unobstructed by defenders

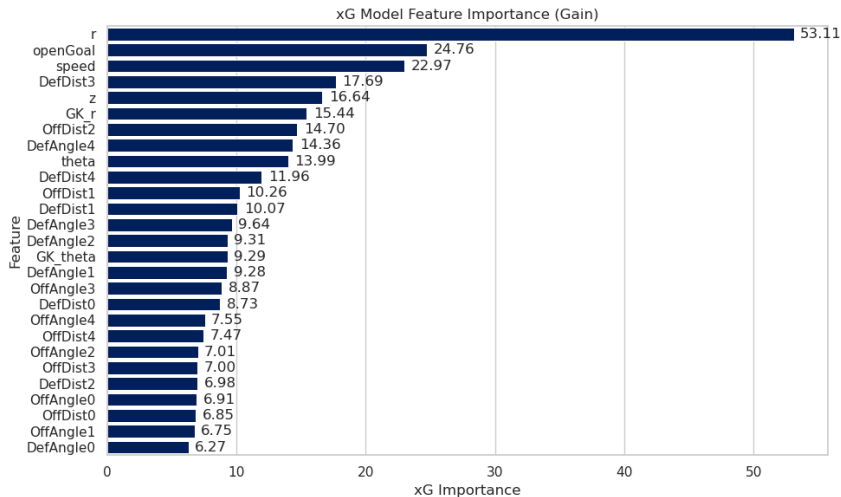
- We train two supervised models using the same feature set:
 - **xS**: $\mathbb{P}(\text{shot in the next second})$
 - **xG**: $\mathbb{P}(\text{goal} \mid \text{shot})$
- Fit using 5-fold cross-validated XGBoost
- Evaluated with log loss (lower is better)

What Drives Shot Creation (xS)?



- Distance and ball speed dominate: proximity and tempo create chances.

What Drives Shot Quality (xG)?



- Distance, openGoal, and ball speed primarily determine shot quality.

Does $xG+$ Predict Goals Better than xG ?

- **Idea:** If a metric measures “goal-scoring danger,” it should predict goals well!
- **Data:** 3 EPL seasons, 114 matchdays.
- **Out-of-sample test:**
 - 1 Hold out one matchday at a time.
 - 2 Train our models on the other 113 matchdays.
 - 3 Predict goals on the held-out matchday.
- **Compare:** xG vs xS vs $xG+$ (with two possession aggregation rules).

Moment-level metrics:

- **xS:** Probability of a shot in the next second
- **xG:** Probability of scoring given a shot
- **xG+:** $xS \times xG$ (probability of scoring in the next second)

Possession aggregation:

- **At-least-one-per-possession:** $1 - \prod(1 - p)$
- **Max-per-possession:** $\max(p)$
- **Baseline:** sum of shot-by-shot xG

Team-Level Results: $xG+ > xS > xG$

Table: Mean Squared Error (MSE) by Metric and Aggregation Method

Aggregation Method	$xG+$	xS	xG
At-least-one-per-possession	2.84	2.90	2.94
Max-per-possession	2.84	2.87	2.91
Sum-of-shots			2.90

- **Takeaway:** Both $xG+$ and xS dominate xG , with $xG+$ as the best overall.

What About Player Ability?

- **Question:** Do we learn more about players from *shot-taking* or *goal-scoring*?
- **Approach:**
 - 1 Aggregate each metric by player and season (xG , xS , $xG+$)
 - 2 Compare to observed goals and shots ($xGOE$, $xSOE$, $xG+OE$)
 - 3 Divide by matches played to get per-game metrics
 - 4 Evaluate year-over-year stability (repeatability)

Key Finding: Finishing Is Not Stable; Shot-Taking Is

Table: Year-to-Year Correlation (Stability) of Each Metric (Per Game)

xGOE	xSOE	xG+OE
0.12	0.63	0.35

- **Interpretation:**

- **Finishing is very noisy:** xGOE is weakly correlated year-to-year (0.12).
- **Shot-taking is repeatable:** xSOE is strongly correlated (0.63).
- **Takeaway:** Top players are consistently great because they *create more shots than expected*, not because they reliably “finish hot.”

Top Overperforming Players

xG+ Over Expected (Per Game)

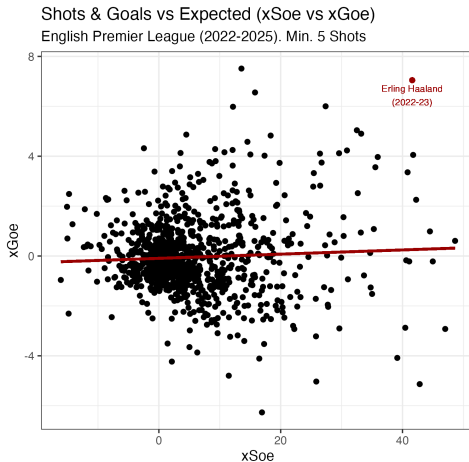
Season	Player	xG+ OE	Goals	Games	Chances
2022-23	Erling Haaland	0.52	36	35	444
2023-24	Erling Haaland	0.44	30	31	513
2024-25	Omar Marmoush	0.41	10	16	293
2024-25	Yoane Wissa	0.35	22	35	429
2024-25	Mohamed Salah	0.33	27	38	1042
2024-25	Chris Wood	0.31	23	36	330
2023-24	Cole Palmer	0.30	19	34	693
2024-25	Alexander Isak	0.30	25	34	558
2023-24	Alexander Isak	0.30	19	28	351
2022-23	Harry Kane	0.29	25	38	589

xG Over Expected (Per Game)

Season	Player	xG OE	Goals	Games	Chances
2024-25	Omar Marmoush	0.41	10	16	293
2024-25	Chris Wood	0.31	23	36	330
2024-25	Michael Keane	0.21	3	10	19
2022-23	Erling Haaland	0.52	36	35	444
2022-23	Roberto Firmino	0.25	10	21	253
2022-23	Martin Ødegaard	0.22	15	37	942
2023-24	Heung-min Son	0.28	20	35	751
2022-23	Matias Viña	0.26	3	10	66
2022-23	Alexander Isak	0.24	11	22	346
2023-24	Taiwo Awoniyi	0.26	7	19	125

Note: Minimum 10 Games with a Chance

Example: Haaland's Record Season



- **Key insight:** Required both taking more shots than expected and favorable finishing variance

- **Best metric for team prediction:** $xG+$
- **Best aggregation method:** at-least-one-per-possession
- **Key insight:** Shot creation is more stable than finishing
- **Why this matters:** match analysis, player evaluation, recruitment, and forecasting
- **Future work:**
 - Sequence models for xS (point processes / hazard models)
 - Hierarchical team/player effects to control for selection bias
 - Value player actions (shoot vs pass vs carry) with counterfactual $xG+$
 - Validate in other leagues and deploy in real time

Questions?



Link to paper, code, and slides

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